

Physics, 2015 EC, entrance exam, version 1

Grade 12 · Physics · 2015 E.C.

Time allowed: 120 minutes

Questions: 46

Total marks: 46

Name _____ Grade _____ ID No. _____ Date _____

1. A good scientific theory is one which is (1)
 - A. simple (less complex), has a wide scope, and is accurate and consistent.
 - B. simple (less complex), specific in scope, and is accurate and consistent.
 - C. more complex, has a wide scope, is accurate and consistent.
 - D. more complex, specific in scope, consistent but not accurate.
2. The number of significant digits in a quantity $m = 7.10 \times 10^{30}$ kg is (1)
 - A. 3
 - B. 2
 - C. 32
 - D. 33
3. Given $A = a\mathbf{i} + b\mathbf{j} + 2\mathbf{k}$ and $B = 4\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}$. What must be the value of 'a' and 'b', respectively, for the two vectors to be parallel? (1)
 - A. -2 and -1
 - B. 2 and 1
 - C. -1 and -2
 - D. 1 and 2
4. For what value of 'a' the two vectors $A = a\mathbf{i} + 2\mathbf{j} + 5\mathbf{k}$ and $B = (1/2)\mathbf{i} + 6\mathbf{j} + 2\mathbf{k}$ are orthogonal to each other? (1)
 - A. 1
 - B. -44
 - C. -1
 - D. 4
5. What is the area of a parallelogram formed by vectors $A = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$ and $B = 3\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$? (1)
 - A. $\sqrt{111}$
 - B. $\sqrt{121}$
 - C. $\sqrt{131}$
 - D. $\sqrt{141}$
6. What is the magnitude of $A \cdot (A \times B)$ if θ is the angle between vector A and vector B? (1)
 - A. $BA \sin\theta$
 - B. $BA^2 \sin\theta$
 - C. $B^2A \sin\theta$
 - D. zero
7. Which one of the following vector properties is WRONG? (1)
 - A. Scalar product is commutative.
 - B. Vector addition is associative.
 - C. Both scalar and vector products are distributive over addition & subtraction.
 - D. The magnitudes of $A \times B$ and $B \times A$ are different.
8. A, B and C are three points, in that order, on a straight line road with $AB=40\text{km}$ and $BC=90\text{km}$. A woman travels from A to B at 10 km/h and then from B to C at 15 km/h. What is the average speed of the woman for the journey from A to C? (1)
 - A. 25 km/h
 - B. 13 km/h
 - C. 5 km/h
 - D. 6 km/h

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9. A stone is thrown upward with a speed of 20 m/s from a point at a height h meters above the horizontal ground level. If the stone hits the ground 5 seconds later, what is the height h ? (1)
- A. zero
B. 100 m
C. 25 m
D. 20 m
-
10. A truck traveling with constant speed of 20 m/s along a straight line passes a motorcycle. Just as the truck passes, the motorcycle accelerates from rest at 2 m/s^2 to overtake the truck. How fast is the motorcycle traveling while it overtakes the truck? (1)
- A. 2 m/s
B. 20 m/s
C. 30 m/s
D. 40 m/s
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11. Which one of the following statements is CORRECT about two dimensional motions? (1)
- A. The acceleration in a uniformly circular motion is constant.
B. In a projectile motion the vertical and horizontal parts of its motion are independent of each other.
C. The maximum range of projectiles can be achieved at an angle of 90° .
D. The velocity of a vertical circular motion is maximum at the top of the vertical circle.
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12. A man of mass 80 kg stands on a balance inside an elevator. If the balance reads 900 N, the elevator is (1)
- A. ascending with constant velocity.
B. descending with constant velocity.
C. ascending with constant acceleration.
D. descending with constant acceleration.
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13. A 32 kg big fish moving with 10 m/s to the east swallows a small 4 kg fish moving with a speed of 5 m/s to the west. The velocity of the big fish after impact is (1)
- A. 8.3 m/s, east
B. 10 m/s, east
C. 10 m/s, west
D. 8.3 m/s, west
-
14. A 10 kg object moving with a velocity of $(6i) \text{ m/s}$ explodes into two equal fragments. One flies off with a velocity $(2i - j) \text{ m/s}$. What is the velocity of the other fragment? (1)
- A. $(3i + 5j) \text{ m/s}$
B. $(-3i - 5j) \text{ m/s}$
C. $(10i + j) \text{ m/s}$
D. $(6i - 5j) \text{ m/s}$
-
15. What should be the coefficient of static friction at the pavement of a flat road for a car moving with a maximum speed of 15 m/s in a circle of radius 45 m to make turns safely? (1)
- A. 0.2
B. 0.3
C. 0.4
D. 0.5
-
16. The dimension of energy is (1)
- A. ML^2T^{-2}
B. MLT^{-2}
C. ML^2T^{-3}
D. MLT^{-3}
-
17. A force of $F = (2i + 4k) \text{ N}$ is exerted on a block to move it with a velocity of $V = (6j + 8k) \text{ m/s}$. What is the power exerted on the block? (1)
- A. 12 W
B. 32 W
C. 24 W
D. 16 W
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18. A block of mass 2.5 kg is attached to a spring with spring constant 40 N/m. If it is pulled down a displacement of 30 cm and released, what would its speed be as it passes the equilibrium position? (1)
- A. 1.6 m/s
B. 1.2 m/s
C. 1.4 m/s
D. 1 m/s

19. What is the moment of inertia of a thin rod of mass M and length L about an axis perpendicular to the rod at a distance of $L/4$ from one end? ($I_{\text{cm}} = \frac{1}{12} ML^2$) (1)
- A. $(7/48)ML^2$ B. $(7/12)ML^2$
C. $(5/12)ML^2$ D. $(1/3)ML^2$
-
20. The angular displacement described by a particle is given as $\theta(t) = 10 + 20t - 4t^2$ where θ in radian and t in seconds. What is the value of t when the particle becomes momentarily at rest? (1)
- A. 5 s B. 2.5 s
C. 2 s D. 1 s
-
21. A solid cylindrical grinding stone ($I = \frac{1}{2}MR^2$) has a radius of 0.4 m and mass of 40 kg. A constant unbalanced torque acting on it changes its angular speed to 40 rad/s from rest in 20 seconds. What is the magnitude of the desired net torque? (1)
- A. 6.4 Nm B. 16 Nm
C. 3.2 Nm D. 8 Nm
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22. A ring ($I = MR^2$) is released and rolled down an inclined plane which is 10 m above the horizontal ground. What is the linear velocity of the ring at the bottom of the incline? (1)
- A. $2\sqrt{10}$ m/s B. $\sqrt{10}$ m/s
C. 100 m/s D. 10 m/s
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23. Real gases behave like an ideal gas at (1)
- A. Low temperature and high pressure. B. High temperature and low pressure.
C. Low temperature and low pressure. D. High temperature and high pressure.
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24. A Carnot engine working between 27°C and 327°C has a work output of 800 J per cycle. The amount of heat energy supplied to the engine from the source in each cycle is (1)
- A. 800 J B. 1600 J
C. 3200 J D. 6400 J
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25. If one mole of an ideal gas doubles its volume as it undergoes an isothermal expansion, its pressure is (1)
- A. quadrupled B. doubled
C. halved D. quartered
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26. The rms speed of an ideal gas at a temperature of 120 K is V . What would the rms speed be if its temperature becomes 480 K? (1)
- A. $V/4$ B. $V/2$
C. $4V$ D. $2V$
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27. A hollow metal sphere of radius 10 cm is charged to a potential of 100 V. What is the potential at the center of the sphere? (1)
- A. Zero B. 50 V
C. 200 V D. 100 V
-
28. A body is said to be in neutral equilibrium if (1)
- A. its center of gravity is above the axis of rotation.
B. its center of gravity is below the axis of rotation.
C. its center of gravity coincides (meet at a point) with the axis of rotation.
D. its center of gravity produces net torque about the axis of rotation.

29. A body is said to be charged positively if it has (1)

- A. excess of protons.
B. deficiency of protons.
C. excess of electrons.
D. deficiency of electrons.

30. A 10^{-6} F capacitor is charged with a 10 V battery. What is the energy of the capacitor? (1)

- A. 50 μJ B. 100 μJ
C. 150 μJ D. 200 μJ

31. The intensity level (loudness level) changes from $L_1 = 60$ dB to $L_2 = 90$ dB. What is the ratio of intensities I_2/I_1 ? (1)

- A. 3 B. 30
C. 300 D. 1000

32. A 0.5 kg object attached to a spring undergoes SHM with amplitude of 8 cm and period of 0.2π s. What is the maximum force acting on the object? (1)

- A. 80 N
B. 50 N
C. 8 N
D. 4 N

33. When light travels from one medium to another which is separated by a sharp boundary the characteristic that does not change is

- A. velocity
B. wavelength
C. frequency
D. amplitude

34. Monochromatic green light of wavelength 5×10^{-7} m illuminates two slits 1 mm apart. If the screen and the slits are 2 m apart, the separation between two successive bright fringes is

- A. 0.25 mm B. 0.1 mm
C. 1 mm D. 0.01 mm

35. A tube closed at one end containing air produces a fundamental frequency of 512 Hz. If the same length of tube is open at both ends, the fundamental frequency that can be produced is

- A. 1024 Hz
B. 512 Hz
C. 256 Hz
D. 125 Hz

36. A uniform spring of force constant K is cut into two equal pieces. What is the force constant of each piece? (1)

- A. $2K$
B. $3K/2$
C. K
D. $K/2$

37. A progressive wave is represented by the equation $Y(t,x) = (0.2) \sin(6.28t - 12.56x)$, where x and y are in meter and t in seconds. The velocity of the wave is (1)

- A. 2 m/s
B. 1.5 m/s
C. 1 m/s
D. 0.5 m/s

38. Which pair of quantities are expressed in the same units? (1)

- A. Stress and strain
B. Young's modulus and stress
C. Young's modulus and strain
D. Shear modulus and strain

39. A 0.5 m long guitar string of cross-sectional area 10^{-6} m^2 has Young's modulus of $Y = 2 \times 10^9 \text{ N/m}^2$. By how much length do you stretch the string to obtain a tension of 20 N? (1)

A. 2 mm
B. 3 mm
C. 4 mm
D. 5 mm

40. Which one of the following statements is WRONG? (1)

A. Wave fronts of point source of light are concentric spheres.
B. Light rays and wave fronts intersect at right angles.
C. Light rays and wave fronts are parallel.
D. Rays and the radii of spherical wave fronts.

41. In a hydraulic brake system a force of 20 N is applied to piston 1 which has a cross-sectional area of 4 cm^2 . What is the force on piston 2 if its cross-sectional area is 60 cm^2 ? (1)

A. 200 N
B. 300 N
C. 400 N
D. 500 N

42. Bernoulli's principle is based on (1)

A. conservation of momentum
B. conservation of mass
C. conservation of energy
D. mass-energy equivalence

43. The speed of water flowing through a horizontal pipe is 10 m/s at a point where the cross-sectional area of the pipe is 0.1 cm^2 . If there is a point in the pipe where its cross-sectional area is 1 cm^2 , what is the pressure difference between these points? (1)

A. $4.95 \times 10^4 \text{ Pa}$
B. $5.95 \times 10^4 \text{ Pa}$
C. $6.95 \times 10^4 \text{ Pa}$
D. $7.95 \times 10^4 \text{ Pa}$

44. The transfer of heat by the direct movement of molecules is called (1)

A. conduction
B. convection
C. radiation
D. induction

45. A window has a glass surface of 2000 cm^2 and thickness of 4 mm. The temperature inside and outside the glass are 21°C and -3°C , respectively. What is the rate of heat flow through the window? ($K = 0.2 \text{ W/m}^\circ\text{C}$) (1)

A. 480 W
B. 48 W
C. 84 W
D. 240 W

46. A piece of iron (specific heat = $450 \text{ J/kg}^\circ\text{C}$) is dropped from a height of 900 m above the ground. Assuming all the energy is converted to heat the iron, what will be the rise in temperature of the iron in degree centigrade? (1)

A. zero
B. 8
C. 20
D. 24

— END OF EXAMINATION —